

Experiment 'p_5_2_a_all_the_same' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_5_2_a_all_the_same

Natural language statement: Write a method returning true if the arguments are all the same.

Method signature: p_5_2_a_all_the_same(x: real, y: real, z: real) returns (all_same: bool)

Ensures

- all_same == (x == y && y == z)

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method returning true if the arguments are all the same.

The signature should be:

`p_5_2_a_all_the_same(x: real, y: real, z: real) returns (all_same: bool)`

The method should respect the following contract:

`ensures all_same == (x == y && y == z)`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmyLXKUfqs8SZa4oa2z9rCTG5erZC

Model: gpt-5.5-2026-04-23

Created at: 1780564087

Finish reason: stop

Usage: 286 tokens in, and 27 tokens out

```
{  
  all_same := x == y && y == z;  
}
```

Final Program

```
method p_5_2_a_all_the_same(x: real, y: real, z: real) returns (all_same: bool)  
  ensures all_same == (x == y && y == z)  
{  
  all_same := x == y && y == z;  
}
```

Total Token Usage

Input tokens: 286

Output tokens: 27

Reasoning tokens: 0

Sum of ‘total tokens’: 313

Experiment Timings

Overall Experiment started at 1780564087414, ended at 1780564126567, lasting 39153ms (39.15 seconds)

Iteration #1 started at 1780564087415, ended at 1780564126566, lasting 39151ms (39.15 seconds)